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SEMICONDUCTOR DEVICE AND SEMICONDUCTOR MODULE

Abstract

A first bonding member is located between a first electrode of a second chip and a second electrode of a first chip in a first direction and between a first electrode of the second chip and a protective film of the first chip in the first direction. The first bonding member electrically connects the second electrode of the first chip and the first electrode of the second chip in an opening of the protective film of the first chip. The protective film of the first chip includes a first recess in a surface of the protective film of the first chip facing the first electrode of the second chip. A portion of the first bonding member is located in the first recess.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-036087, filed on Mar. 8, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a semiconductor device and a semiconductor module.

BACKGROUND

[0003] IGBTs (Insulated Gate Bipolar Transistors) are widely used as power semiconductor devices that control high breakdown voltages and large currents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** is a schematic cross-sectional view of a first chip and a second chip of a semiconductor device of an embodiment;

[0005] FIG. **2** is a schematic plan view of the first chip and the second chip of the embodiment;

[0006] FIG. **3** is an equivalent circuit diagram of the semiconductor device of the embodiment;

[0007] FIG. **4** is a schematic plan view of a semiconductor module of an embodiment;

[0008] FIG. **5** is a schematic cross-sectional view of the semiconductor module of the embodiment; and

[0009] FIG. **6** is a schematic cross-sectional view of a first chip and a second chip according to a modified example of the embodiment.

DETAILED DESCRIPTION

[0010] According to one embodiment, a semiconductor device includes a first chip and a second chip stacked in a first direction; and a first bonding member located between the first chip and the second chip in the first direction, the first chip and the second chip each including a semiconductor part including a first surface, and a second surface positioned at a side opposite to the first surface in the first direction, a first electrode located at the first surface, a second electrode located at a portion of the second surface, and a protective film partially covering the second surface, the protective film including an opening that exposes the second electrode, the first bonding member being located between the first electrode of the second chip and the second electrode of the first chip in the first direction and between the first electrode of the second chip and the protective film of the first chip in the first direction, the first bonding member electrically connecting the second electrode of the first chip and the first electrode of the second chip in the opening of the protective film of the first chip, the protective film of the first chip including a first recess in a surface of the protective film of the first chip facing the first electrode of the second chip, a portion of the first bonding member being located in the first recess.

[0011] Exemplary embodiments will now be described with reference to the drawings.

[0012] The drawings are schematic or conceptual; and the relationships between the thickness and width of portions, the proportional coefficients of sizes among portions, etc., are not necessarily the same as the actual values thereof. Furthermore, the dimensions and proportional coefficients may be illustrated differently among drawings, even for identical portions.

[0013] In the specification of the application and the drawings, components similar to those described in regard to a drawing thereinabove are marked with like reference numerals, and a detailed description is omitted as appropriate.

[0014] A semiconductor device **100** of an embodiment will now be described with reference to FIGS. **1** to **3**.

[0015] The semiconductor device **100** includes a first chip **101**, a second chip **102**, and a first bonding member **51**. As shown in FIG. **1**, the first chip **101** and the second chip **102** are stacked in a first direction Z. A second electrode **22** of the first chip **101** and the second electrode **22** of the second chip **102** are positioned to overlap each other in the first direction Z. The first bonding member **51** is located between the first chip **101** and the second chip **102** in the first direction Z.

[0016] The first chip **101** and the second chip **102** are similarly configured. The first chip **101** and the second chip **102** each include a semiconductor part **10**, a first electrode **21**, the second electrode **22**, and a protective film **40**.

[0017] The semiconductor part **10** includes a first surface **10**A, and a second surface **10**B positioned at the side opposite to the first surface **10**A in the first direction Z. The first electrode **21** is located at the first surface **10**A. The second electrode **22** is located at a portion of the second surface **10**B.

[0018] For example, the semiconductor part **10** is made of silicon carbide (SiC). In the description of the specification, a first conductivity type is taken to be an n-type; and a second conductivity type is taken to be a p-type. Or, the first conductivity type may be the p-type; and the second conductivity type may be the n-type.

[0019] The semiconductor part **10** includes an n-type first semiconductor layer **11** located on the first electrode **21**, an n-type second semiconductor layer **12** located on the first semiconductor layer **11**, and a p-type third semiconductor layer **13** located on the second semiconductor layer **12**. The n-type impurity concentration of the second semiconductor layer **12** is less than the n-type impurity concentration of the first semiconductor layer **11**. Conversely speaking, the n-type impurity concentration of the first semiconductor layer **11** is greater than the n-type impurity concentration of the second semiconductor layer **12**. The thickness in the first direction Z of the second semiconductor layer **12** is greater than the thickness in the first direction Z of the first semiconductor layer **11**. The first semiconductor layer **11** is electrically connected with the first electrode **21**. The second electrode **22** is located on the third semiconductor layer **13**; and the third semiconductor layer **13** is electrically connected with the second electrode **22**.

[0020] For example, the first chip **101** and the second chip **102** each are diode chips. The first electrode **21** functions as a cathode electrode; and the second electrode **22** functions as an anode electrode. A current flows in the first direction Z between the second electrode **22** and the first electrode **21** in each of the first and second chips **101** and **102**.

[0021] The protective film **40** partially covers the second surface **10**B of the semiconductor part **10**. The protective film **40** includes an opening **43** that exposes the second electrode **22**. The protective film **40** is an insulating film, and is, for example, a resin film. The protective film **40** includes, for example, mainly polyimide.

[0022] The first bonding member **51** is conductive. The first bonding member **51** is located between the first electrode **21** of the second chip **102** and the second electrode **22** of the first chip **101** in the first direction Z and between the first electrode **21** of the second chip **102** and the protective film **40** of the first chip **101** in the first direction Z. The first bonding member **51** electrically connects the second electrode **22** of the first chip **101** and the first electrode **21** of the second chip **102** in the opening **43** of the protective film **40** of the first chip **101**. The first chip **101** and the second chip **102** are connected in series via the first bonding member **51**. For example, solder, silver, etc., can be used as the material of the first bonding member **51**.

[0023] The protective film **40** of the first chip **101** includes a first recess **41** in the surface of the

protective film **40** of the first chip **101** facing the first electrode **21** of the second chip **102**. A portion of the first bonding member **51** is located in the first recess **41**. By including the first recess **41**, the volume of the first bonding member **51** that can be held between the first chip **101** and the second chip **102** stacked in the first direction Z is increased, and the first bonding member **51** does not easily flow out onto the side surface of the first chip **101** and the side surface of the second chip **102**. As a result, unintended short-circuit defects can be suppressed, and a semiconductor device with high reliability can be provided.
[0024] As shown in FIG. **2**, it is favorable for the first recess **41** to be a trench continuously surrounding the second electrode **22** when the second surface **10**B is viewed in plan. As a result, the first bonding member **51** does not easily flow out onto the side surface of the first chip **101** and the side surface of the second chip **102** in all directions around the second electrode **22**. The protective film **40** of the first chip **101** contacts the first bonding member **51** in at least a portion of the region inward of the first recess **41**. The protective film **40** of the first chip **101** may contact the first bonding member **51** in the region outward of the first recess **41**.
[0025] As shown in FIG. **2**, the second surface **10**B of the semiconductor part **10** can include an element region R**1** in which the p-type third semiconductor layer **13** is positioned, and a termination region R**2** positioned outward of the element region R**1**. The termination region R**2** continuously surrounds the element region R**1**. In FIG. **2**, an outer edge **13**A of the third semiconductor layer **13** can be the boundary between the element region R**1** and the termination region R**2**.
[0026] In the termination region R**2** as shown in FIG. **1**, a p-type fourth semiconductor layer **14** that has a lower p-type impurity concentration than the third semiconductor layer **13** is positioned on the second semiconductor layer **12**. The fourth semiconductor layer **14** continuously surrounds the third semiconductor layer **13** when viewed in plan. The portion of the fourth semiconductor layer **14** at the inner perimeter side contacts the portion of the third semiconductor layer **13** at the outer perimeter side. The p-n junction between the third semiconductor layer **13** and the second semiconductor layer **12** is positioned inward of the inner edge of the fourth semiconductor layer **14**. The potential of the second electrode **22** is applied to the fourth semiconductor layer **14** via the third semiconductor layer **13**. Due to the fourth semiconductor layer **14** that has a lower p-type impurity concentration than the third semiconductor layer **13**, the depletion layer can easily extend into the termination region R**2**; and the breakdown voltages of the first and second chips **101** and **102** can be increased.
[0027] The depth of the first recess **41** is less than the thickness of the protective film **40** on the termination region R**2**, and so reliability degradation due to a partial thickness reduction of the protective film **40** on the termination region R**2** can be suppressed. The depth and the thickness are lengths in the first direction Z. The first recess **41** has a depth of not less than 100 nm and not more than 20,000 nm and a width of not less than 100 nm.
[0028] As shown in FIG. **1**, the first chip **101** and the second chip **102** each may further include an insulating film **30** located between the protective film **40** and the second surface **10**B of the semiconductor part **10**. The insulating film **30** is, for example, an inorganic film. The insulating film **30** is, for example, a silicon oxide film. The film thickness of the protective film **40** is, for example, several tens of times to several hundreds of times the film thickness of the insulating film **30**.
[0029] As shown in FIG. **3**, the semiconductor device **100** of the embodiment can further include a third chip **103** that is connected in parallel to the first and second chips **101** and **102**.
[0030] The third chip **103** is a transistor chip that includes a third electrode **103**C electrically connected with the first electrode **21** of the first chip **101**, and a fourth electrode **103**E electrically connected with the second electrode **22** of the second chip **102**. The third chip **103** is, for example, an IGBT (Insulated Gate Bipolar Transistor) chip. In the IGBT chip, the third electrode **103**C is the collector electrode; and the fourth electrode **103**E is the emitter electrode. The third chip **103**

further includes a gate electrode **103**G. The third chip **103** is, for example, an IGBT chip of a silicon material.

[0031] For example, the first chip **101** and the second chip **102** function as freewheeling diodes that carry a reverse current generated when the third chip **103** (the IGBT chip) switches in applications such as inverters, motor driving, etc. The breakdown voltage of the third chip **103** (the IGBT chip) is, for example, 4.5 kV. In the freewheeling diode of the embodiment, for example, a freewheeling diode that has a breakdown voltage of 4.5 kV is realized by using a configuration in which the first chip **101** (or the second chip **102**) that has a breakdown voltage of 1.2 kV and the second chip **102** (or the first chip **101**) that has a breakdown voltage of 3.3 kV are connected in series. As a result, the cost can be less than when one diode chip having a breakdown voltage of 4.5 kV is used, and the reliability can be increased.

[0032] A semiconductor module **200** of an embodiment will now be described with reference to FIGS. **4** and **5**.

[0033] As shown in FIG. **5**, the semiconductor module **200** includes a first conductive member **201**, a second conductive member **202**, and a plurality of the aforementioned semiconductor devices **100** located between the first conductive member **201** and the second conductive member **202** in the first direction Z. The multiple semiconductor devices **100** each include the first to third chips **101** to **103**. In other words, the semiconductor module **200** includes the multiple first chips **101**, the multiple second chips **102**, and the multiple third chips **103**. The multiple first chips **101** and the multiple third chips **103** each are separated in a plane crossing the first direction Z. As described above, the first chip **101** and the second chip **102** are stacked with the first bonding member **51** interposed, and are connected in series. In FIG. **4**, the first chip **101** and the second chip **102** that are stacked are illustrated by cross hatching.

[0034] The first electrode **21** (the cathode electrode) of the first chip **101** and the third electrode **103**C (the collector electrode) of the third chip **103** are electrically connected with the first conductive member **201**. A first metal plate **205** is located between the first chip **101** and the first conductive member **201**; and the first conductive member **201** and the first electrode **21** of the first chip **101** are electrically connected via the first metal plate **205**. A third metal plate **203** is located between the third chip **103** and the first conductive member **201**; and the first conductive member **201** and the third electrode **103**C of the third chip **103** are electrically connected via the third metal plate **203**.

[0035] The second electrode **22** (the anode electrode) of the second chip **102** and the fourth electrode **103**E (the emitter electrode) of the third chip **103** are electrically connected with the second conductive member **202**. A second metal plate **206** is located between the second chip **102** and the second conductive member **202**; and the second conductive member **202** and the second electrode **22** of the second chip **102** are electrically connected via the second metal plate **206**. A fourth metal plate **204** is located between the third chip **103** and the second conductive member **202**; and the second conductive member **202** and the fourth electrode **103**E of the third chip **103** are electrically connected via the fourth metal plate **204**.

[0036] For example, copper can be used as the materials of the first and second conductive members **201** and **202**. The first conductive member **201** and the second conductive member **202** extend along a plane crossing the first direction Z. The first conductive member **201** and the second conductive member **202** include multiple protrusions extending along the first direction Z. The multiple protrusions contact the first metal plate **205**, the second metal plate **206**, the third metal plate **203**, and the fourth metal plate **204**. For example, molybdenum can be used as the materials of the first metal plate **205**, the second metal plate **206**, the third metal plate **203**, and the fourth metal plate **204**.

[0037] The semiconductor module **200** can further include a wiring member **207** that is located between the first conductive member **201** and the second conductive member **202** in the first direction Z and is electrically connected with the gate electrode **103**G of the third chip **103**. The gate electrode **103**G of the third chip **103** is electrically connected with a wiring layer formed in the wiring member **207** via a connection member **208**. For example, a spring pin can be used as the connection member **208**. The third chip **103** is switched on and off by a gate voltage applied to the gate electrode **103**G via the wiring member **207**.

[0038] For example, the state in which the first to third chips **101** to **103** are clamped between the first conductive member **201** and the second conductive member **202** is maintained by the fastening force of a screw connected to the first and second conductive members **201** and **202**. Pressure along the first direction Z is applied to the first to third chips **101** to **103**.

[0039] The semiconductor module **200** further includes a second bonding member **52** (shown in FIG. **1**) that is located between the second conductive member **202** and the second chip **102** and electrically connects the second conductive member **202** and the second electrode **22** of the second chip **102**. A material similar to that of the first bonding member **51** can be used as the material of the second bonding member **52**.

[0040] The protective film **40** of the second chip **102** includes a second recess **42** in the surface (the upper surface in FIG. **1**) of the protective film **40** of the second chip **102** facing the second conductive member **202**. A portion of the second bonding member **52** is located in the second recess **42**.

[0041] As the operation of switching the third chip **103** on and off is repeated, the first bonding member **51** tends to flow due to the repeating thermal expansion and contraction between the first chip **101** and the second chip **102**. Similarly, the second bonding member **52** tends to flow due to the repeating thermal expansion and contraction between the second conductive member **202** and the second chip **102**.

[0042] According to the embodiment above, by including the first recess **41**, the first bonding member **51** does not easily flow out onto the side surface of the first chip **101** and the side surface of the second chip **102** even when the first bonding member **51** flows easily. Similarly, by including the second recess **42**, the second bonding member **52** does not easily flow out onto the side surface of the second chip **102** and the side surface of the first chip **101**. As a result, unintended short-circuit defects can be suppressed, and a semiconductor module with high reliability can be provided.

[0043] Similarly to the first recess **41** of the first chip **101**, it is favorable for the second recess **42** to be a trench continuously surrounding the second electrode **22** of the second chip **102** when the second surface **10**B of the second chip **102** is viewed in plan.

[0044] The second surface **10**B of the semiconductor part **10** may be flat, as in a modification of the first chip **101** (or the second chip **102**) shown in FIG. **6**. Also, multiple third semiconductor layers **13** that are arranged to be separated from each other in a plane crossing the first direction Z may be included. The second electrode **22** may have a stacked structure of a first layer **22**A, a second layer **22**B, and a third layer **22**C. The first layer **22**A is located on the second surface **10**B of the semiconductor part **10** and contacts the third semiconductor layer **13**. The first layer **22**A includes, for example, Ni. The second layer **22**B is located on the first layer **22**A and includes, for example, Ti. The third layer **22**C is located on the second layer **22**B and includes, for example, Au.

[0045] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modification as would fall within the scope and spirit of the inventions.

Claims

1. A semiconductor device, comprising: a first chip and a second chip stacked in a first direction; and a first bonding member located between the first chip and the second chip in the first direction, the first chip and the second chip each including a semiconductor part including a first surface, and a second surface positioned at a side opposite to the first surface in the first direction, a first electrode located at the first surface, a second electrode located at a portion of the second surface, and a protective film partially covering the second surface, the protective film including an opening that exposes the second electrode, the first bonding member being located between the first electrode of the second chip and the second electrode of the first chip in the first direction and between the first electrode of the second chip and the protective film of the first chip in the first direction, the first bonding member electrically connecting the second electrode of the first chip and the first electrode of the second chip in the opening of the protective film of the first chip, the protective film of the first chip including a first recess in a surface of the protective film of the first chip facing the first electrode of the second chip, a portion of the first bonding member being located in the first recess.
2. The semiconductor device according to claim 1, wherein the first recess is a trench continuously surrounding the second electrode when the second surface is viewed in plan.
3. The semiconductor device according to claim 1, wherein the first chip and the second chip each are diode chips, and the semiconductor part includes: a first semiconductor layer located on the first electrode, the first semiconductor layer being of a first conductivity type; a second semiconductor layer located on the first semiconductor layer, the second semiconductor layer being of the first conductivity type and having a lower first-conductivity-type impurity concentration than the first semiconductor layer; and a third semiconductor layer located on the second semiconductor layer, the third semiconductor layer being electrically connected with the second electrode, the third semiconductor layer being of a second conductivity type.
4. The semiconductor device according to claim 3, further comprising: a third chip connected in parallel to the first and second chips, the third chip being a transistor chip including a third electrode electrically connected with the first electrode of the first chip, and a fourth electrode electrically connected with the second electrode of the second chip.
5. The semiconductor device according to claim 4, wherein the third chip is an IGBT (Insulated Gate Bipolar Transistor) chip.
6. The semiconductor device according to claim 3, wherein the second surface of the semiconductor part includes: an element region in which the third semiconductor layer is positioned; and a termination region positioned outward of the element region, the semiconductor part further includes a fourth semiconductor layer positioned on the second semiconductor layer in the termination region, and the fourth semiconductor layer is of the second conductivity type and has a lower second-conductivity-type impurity concentration than the third semiconductor layer.
7. The semiconductor device according to claim 1, wherein the semiconductor part is made of silicon carbide.
8. A semiconductor module, comprising: a plurality of the semiconductor devices according to claim 4; a first conductive member; and a second conductive member, the plurality of semiconductor devices being located between the first conductive member and the second conductive member in the first direction, the first electrode of the first chip and the third electrode of the third chip being electrically connected with the first conductive member, the second electrode of the second chip and the fourth electrode of the third chip being electrically connected with the second conductive member.
9. The semiconductor module according to claim 8, further comprising: a second bonding member located between the second conductive member and the second chip, the second bonding member electrically connecting the second conductive member and the second electrode of the second chip.
10. The semiconductor module according to claim 9, wherein the protective film of the second chip

includes a second recess in a surface of the protective film of the second chip facing the second conductive member, and a portion of the second bonding member is located in the second recess.

11. The semiconductor module according to claim 10, wherein the second recess is a trench continuously surrounding the second electrode of the second chip when the second surface of the second chip is viewed in plan.

12. The semiconductor module according to claim 8, further comprising: a wiring member located between the first conductive member and the second conductive member in the first direction, the wiring member being electrically connected with a gate electrode of the third chip.